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MARSNet: A convolutional attention residual shrinkage network for RNA-protein binding site prediction

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ABSTRACT

RNA-binding proteins (RBPs) regulate post-transcriptional gene expression by recognizing specific RNA elements, making accurate identification of RBP binding sites essential for understanding gene regulation and disease mechanisms. However, experimental profiling is costly and often noisy, motivating robust sequence-based computational prediction. We propose **MARSNet**, a convolutional attention residual shrinkage network for RNA–protein binding site prediction. MARSNet adopts sequential multi-scale window encoding to capture both local motifs and broader contextual dependencies. For each window scale, a feature extractor integrates residual shrinkage blocks with lightweight attention modules. The residual shrinkage mechanism performs learnable soft-thresholding to suppress low-magnitude activations induced by background noise, significantly improving robustness, especially on data-scarce or heterogeneous datasets. Predictions from different window scales are then aggregated by a weighted fusion strategy to form the final output. Extensive experiments on the RBP-24 benchmark demonstrate that MARSNet achieves an average AUROC of **0.957**, AP of **0.875**, and MCC of **0.821**, outperforming classic baselines and remaining competitive with recent methods such as PIONet and RMDNet. Notably, MARSNet exhibits superior performance on small-scale datasets, highlighting its stability. Furthermore, interpretability analyses, including motif discovery and an *in silico* mutagenesis case study on the *TARDBP* gene, confirm that MARSNet captures biologically meaningful binding determinants and is sensitive to pathogenic mutations. The source code is available at https://github.com/HNUBioinformatics/MARSNet.

1. Introduction

RNA-binding proteins (RBPs) regulate a wide range of post-transcriptional processes, including alternative splicing, mRNA localization, stability, and translation, thereby shaping gene expression programs in a context-dependent manner (Gerstberger et al., 2014; Seufert et al., 2022; Wang et al., 2008). Aberrant RBP expression or dysregulated RNA–protein interactions have been implicated in diverse diseases (Cookson, 2017; Lukong et al., 2008). For instance, abnormal upregulation of the RBP EWSR1 has been reported to promote oncogenic phenotypes and may contribute to the progression of multiple myeloma (Nishiyama et al., 2021). Therefore, accurately identifying RNA–protein binding sites is essential for understanding disease mechanisms and for developing targeted therapeutic strategies.

Experimentally, genome-wide RBP binding sites can be profiled by CLIP-based technologies, such as crosslinking and immunoprecipitation (CLIP) (Hafner et al., 2021; Ule et al., 2003), high-throughput sequencing of RNA isolated by CLIP (HITS-CLIP) (Chi et al., 2009), and photoactivatable-ribonucleoside-enhanced CLIP (PAR-CLIP) (Ascano et al., 2012; Hafner et al., 2010a,b). Despite their effectiveness, these assays remain time-consuming, computationally demanding, and costly, motivating the development of computational predictors. Early machine learning approaches typically relied on handcrafted sequence/structure features. For example, GraphProt employs support vector machines (SVMs) with graph-kernel representations to integrate sequence and secondary-structure information (Maticzka et al., 2014). IONMF applies orthogonal nonnegative matrix factorization to integrate heterogeneous data sources for improved binding site prediction

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(Meng et al., 2018; Stražar et al., 2016). While these methods are often efficient and interpretable, their representational capacity can be limited when binding determinants involve complex and context-dependent patterns.

Recently, deep learning has been widely applied in speech recognition, medical diagnosis, natural language processing, and image recognition (Cheng et al., 2023). Motivated by its ability to learn rich representations, many deep learning approaches have also been developed for predicting RNA-binding protein (RBP) binding sites. For example, kDeepBind predicts RNA–protein binding sites using convolutional neural networks (CNNs) together with k-gram features (Tahir et al., 2021). SA-Net introduces a k-mer embedding layer to represent subsequences and then applies a self-attention–based model to highlight binding-related patterns (Wang et al., 2022). iDeepE uses CNNs to capture both local and global sequence features for binding site prediction (Pan & Shen, 2018), and MCNN further extends this idea by extracting multiple scales of global and local features with parallel CNN branches (Pan et al., 2022). iDeep combines CNNs with deep belief networks to predict binding sites and infer binding motifs (Pan & Shen, 2017; Tang et al., 2016). DeepRPK encodes sequence and structure features with embedding methods and then applies a CNN and a bidirectional long short-term memory network (BiLSTM) to learn higher-level representations (Deng et al., 2020). Deepnet-RBP processes sequence and structure information with a deep belief network to identify binding sites (Zhang et al., 2016). More recently, CRMSNet integrates multi-head self-attention, CNNs, and recurrent networks; its residual attention block combines residual connections with multi-head self-attention and convolution to improve prediction performance (He et al., 2016; Pan et al., 2023; Targ et al., 2016; Vaswani et al., 2017).

Despite these advances, deep learning models still face practical challenges compared with conventional machine learning methods. First, CLIP-derived sequences often include experimental noise and artifacts, which can introduce spurious signals and reduce generalization. Second, some RBPs have limited labeled data, making complex models prone to overfitting. Third, strong prediction accuracy alone is not sufficient: additional analyses are often required to relate model outputs to known binding preferences and motifs. Finally, deep models typically require more computation, which can hinder large-scale use.

These considerations motivate architectures that are both robust and efficient. Related work on robust neural network design in control and state estimation (Cao et al., 2024; Chandrasekar et al., 2025) further supports the value of stability under noisy conditions. In this study, we therefore develop a lightweight framework that suppresses noisy activations via residual shrinkage and enhances informative features using attention mechanisms, using lightweight attention mechanisms (ECA and CBAM).

To address the above issues, we propose **MARSNet**, a convolutional attention residual shrinkage network for RNA–protein binding site prediction. The key strategy is to combine (i) residual shrinkage with soft-thresholding to suppress noise-induced activations, (ii) lightweight attention mechanisms to enhance informative features, and (iii) multi-scale context modeling to capture binding signals across different receptive fields. Specifically, we design a convolutional attention residual shrinkage module (**ARSNet**) as the basic building block. ARSNet integrates residual shrinkage units with dual attention mechanisms—Efficient Channel Attention (ECA) and Convolutional Block Attention Module (CBAM)—to refine feature representations. Furthermore, we introduce sequential multi-scale window coding and a weighted fusion strategy to aggregate predictions from multiple window sizes (101/201/301/401), mitigating the limitations of single-scale views. In addition, we explore a heterogeneous fusion variant (**MARSNet-F**) to analyze backbone compatibility, finding that fusing optimized homogeneous backbones yields superior performance compared to naïve averaging of heterogeneous models.

Note on Configuration: Unless otherwise specified, MARSNet refers to the final configuration with $N = 4$ fusion groups/stages. This parameter N represents the depth of the multi-scale fusion integration and is varied principally in the ablation studies to demonstrate the saturation of model performance.

Our main contributions are summarized as follows:

- The proposed MARSNet framework introduces a residual shrinkage mechanism and dual attention modules to effectively suppress noise and capture context-dependent binding features.
- Extensive evaluations on the RBP-24 benchmark show that MARSNet achieves state-of-the-art performance, outperforming recent baselines and demonstrating exceptional robustness on small-scale datasets.
- We validate the biological relevance of the model through motif discovery and an *in silico* mutagenesis case study on the *TARDBP* gene, confirming its capability to identify pathogenic mutations and functional binding determinants.

2. Materials and methods

2.1. Datasets and benchmark protocol

We evaluate MARSNet on the widely used RBP-24 benchmark, which contains 24 RBP datasets and is publicly available at http://www.bioinf.uni-freiburg.de/Software/GraphProt. Each dataset provides four files: positive training sequences, negative training sequences, positive test sequences, and negative test sequences. Among the 24 datasets, PTB was generated using HITS-CLIP (Xue et al., 2009), while the remaining 23 datasets were generated using PAR-CLIP and were originally curated from the doRiNA database (Anders et al., 2012). Positive sequences correspond to subsequences centered at CLIP peaks and thus contain experimentally supported binding-site signals, whereas negative sequences are generated by the benchmark protocol and contain no evidence of binding events.

For model development, we strictly follow the predefined training/test splits provided by the GraphProt benchmark to ensure comparability with prior work.

To ensure the generalizability of the model and mitigate overfitting due to redundant sequences, we follow the standard preprocessing protocol of the RBP-24 benchmark. Specifically, redundant sequences were removed using CD-HIT with a similarity threshold of 0.8. In addition to the benchmark-provided redundancy removal, we performed an explicit train–test redundancy/leakage check using 8-mer Jaccard similarity to quantify potential overlap between the predefined splits. The summary statistics and similarity distribution are reported in Table S1 and Fig. S1, respectively. We further conducted a threshold-based filtering sensitivity analysis (Table S2), and dummyTXdummy-(observed negligible changes in AUC/AP. The training sample sizes for all datasets are summarized in Table 1. This preprocessing reduces redundancy and helps mitigate homology bias.

Unless otherwise specified by the benchmark, each test set contains 500 positive and 500 negative sequences; notably, ALKBH5 and C17ORF85 have smaller test sets following the original benchmark setting.

2.2. Sequence preprocessing and multi-scale windowing

Each RNA sequence is represented using a 4-letter alphabet {A, C, G, U}. In implementation, some tools use a DNA alphabet; in such cases, U can be treated as the 4th channel (equivalent to T) without changing the one-hot representation.

Given an input sequence of length L, we apply a multi-scale window strategy to capture both local binding motifs and broader contextual signals. The selection of window sizes is motivated by biological considerations: smaller windows focus on local sequence motifs (typically 4–8 nt), while larger windows capture long-range dependencies and

Table 1
Training sample sizes for the 24 datasets in the RBP-24 benchmark.

RBP	Positive	Negative	RBP	Positive	Negative
ALKBH5	1347	1329	ELAVL1B	27,775	24,474
C17ORF85	2066	2054	ELAVL1A	9964	9783
C22ORF28	9869	9636	EWSR1	16,792	15,220
CAPRIN1	8640	8401	FUS	35,081	31,980
AGO2	48,595	44,751	ELAVL1C	125702	114186
ELAVL1H	9095	8936	IGF2BP1–3	9039	7338
SRSF1	19,938	17,695	MOV10	14,293	13,487
HNRNPC	21,927	20,294	PUM2	9616	8727
TDP-43	92,531	75,579	QKI	10,776	9642
TIA1	18,549	16,635	TAF15	7798	7106
TIAL1	42,832	37,152	PTB	45,074	44,200
AGO1–4	37,402	31,810	ZC3H7B	21,462	20,518

secondary structure contexts. We use window lengths:

$$W_i = W_{\min} + 100(i-1), \quad i = 1, \dots, t, \quad t = \left\lfloor \frac{W_{\max} - W_{\min}}{100} \right\rfloor + 1, \tag{1}$$

where $W_{\min} = 101$ and $W_{\max} = 401$, resulting in $t = 4$ window sizes: $\{101, 201, 301, 401\}$.

If $L \le W$, the sequence is padded with an ambiguous nucleotide token N to length W. If $L > W$, we split the sequence into overlapping subsequences of length W with overlap size $O = 50$:

$$s^{(j)} = s\left[(j-1)(W-O)+1 : (j-1)(W-O)+W\right], \quad j = 1, \dots, C_W, \tag{2}$$

where C_W is the number of sub-windows. *Implementation detail:* to enable efficient batching, we cap/pad each bag to a predefined maximum channel number (consistent with $W_{\max} = 401$ and $O = 50$), i.e., $C_{101} = 7$, $C_{201} = 3$, $C_{301} = 2$, and $C_{401} = 1$. If fewer sub-windows are generated, we pad the remaining channels with all-N sequences.

We then apply one-hot encoding to each windowed sequence and obtain an input tensor $X \in \mathbb{R}^{C_W \times (W+2m) \times 4}$, where $m = 3$ is a boundary padding length to improve convolutional motif scanning near sequence ends. For each position x, the one-hot vector is defined as follows, with ambiguous nucleotides (N) encoded uniformly:

$$M_x = \begin{cases} [1,0,0,0], & s_x = A \\ [0,1,0,0], & s_x = C \\ [0,0,1,0], & s_x = G \\ [0,0,0,1], & s_x = U \\ [0.25,0.25,0.25,0.25], & s_x = N \end{cases}. \tag{3}$$

2.3. The MARSNet architecture

MARSNet is designed as a multi-scale, multi-stage fusion framework for RBP binding site prediction (Fig. 1a). The framework consists of two main components: the overall multi-stage fusion strategy and the core ARSNet feature extractor.

2.3.1. Overview of multi-scale fusion

MARSNet consists of N fusion groups/stages. Based on the empirical ablation study (detailed in Section 3.2.1), we set $N = 4$ in the final model configuration. Within each group, we train one ARSNet model per window size W_i and fuse their predictions via a weighted average:

$$\hat{y}^{(g)} = \sum_{i=1}^{t} \alpha_i \, \hat{y}^{(g)}_{W_i}, \quad \sum_{i=1}^{t} \alpha_i = 1, \tag{4}$$

where $\hat{y}^{(g)}_{W_i}$ is the prediction score produced by ARSNet at window size W_i in group g. In our implementation, we use fixed weights $\boldsymbol{\alpha} = [0.1, 0.2, 0.4, 0.3]$. Finally, MARSNet outputs the average prediction over groups:

$$\hat{y} = \frac{1}{N} \sum_{g=1}^{N} \hat{y}^{(g)}. \tag{5}$$

2.3.2. ARSNet: The core feature extractor

ARSNet is the base learner used in MARSNet (Fig. 1b). Given the encoded input tensor, ARSNet first applies 2D convolutions to learn local motif-like patterns, followed by Batch Normalization (BN). This is followed by two key components designed to enhance feature robustness: the Residual Shrinkage Block Unit (RSBU) and Convolutional Attention Modules.

Residual shrinkage for noise suppression. The RSBU introduces a learnable soft-thresholding operation into the residual branch (Fig. 1c). Weak activations are more likely to be caused by noise, while strong activations reflect true binding patterns. The soft-thresholding function transforms the input feature x to output y as follows:

$$y = \begin{cases} x - \tau & x > \tau \\ 0 & -\tau \le x \le \tau \\ x + \tau & x < -\tau \end{cases}, \tag{6}$$

where $\tau > 0$ is the learned threshold parameter. This effectively forces noise-related features to zero.

Convolutional attention modules. To further emphasize informative features, we integrate two attention mechanisms:

- **Efficient Channel Attention (ECA):** A lightweight module using 1D convolution to model local cross-channel interactions (Fig. 1d): $\rho = \sigma(\mathrm{Conv1D}_k(q))$.
- **Convolutional Block Attention Module (CBAM):** A module that sequentially applies channel and spatial attention (Fig. 1e) to refine the feature maps.

2.4. Multi-network fusion baseline: MARSNet-F

To further analyze fusion behavior, we implement MARSNet-F, which fuses predictions from heterogeneous architectures (ARSNet, CNN, SACNN-BiLSTM, SACNN-BiGRU) via simple averaging:

$$\hat{y}_{\text{MARSNet-F}} = \frac{1}{K} \sum_{k=1}^{K} \hat{y}_k, \tag{7}$$

where $K = 4$. This baseline helps clarify the advantage of our homogeneous multi-scale fusion strategy. As illustrated in Fig. 2, MARSNet-F performs a simple averaging over heterogeneous backbones.

2.5. Implementation and evaluation

Implementation Details: All models are implemented in PyTorch. We use RMSprop optimization with a learning rate of 1×10^{-3} and weight decay of 1×10^{-4}. The batch size is set to 128, and each model is trained for 50 epochs. To facilitate fair computational comparisons, we report the number of parameters and training/inference time measured under the same hardware environment. We also monitor GPU memory usage to confirm that all models can be trained on a single NVIDIA RTX 3080.

Evaluation Metrics: We evaluate prediction performance using the Area Under the Receiver Operating Characteristic curve (AUROC), Average Precision (AP), accuracy (Acc), precision, recall, F1-score, and Matthews Correlation Coefficient (MCC). AUROC is threshold-independent, while MCC provides a balanced measure for binary classification.

3. Results and discussion

3.1. Experimental setup and hyperparameter settings

We followed common practice in RBP binding site prediction and used the same predefined training/test splits of the RBP-24 benchmark. Unless otherwise stated, all models were implemented in PyTorch (v1.8.1) and trained on a single NVIDIA RTX 3080 GPU. The input sequences were encoded using the proposed multi-scale windows

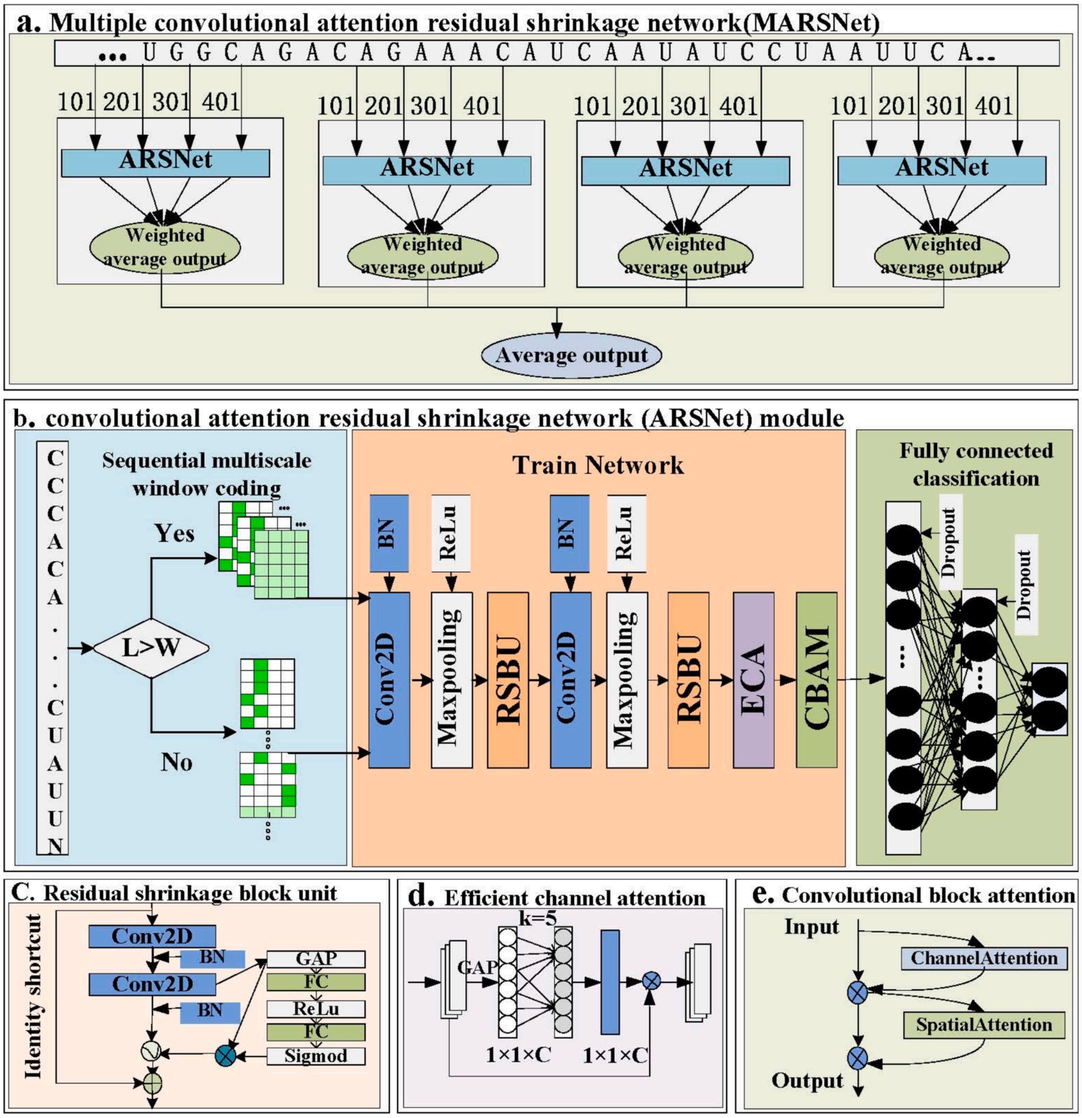

Fig. 1. Overview of the MARSNet framework and its components. (a) The MARSNet architecture with multi-scale windows and repeated fusion. (b) The ARSNet module as the base unit. (c) The Residual Shrinkage Block Unit (RSBU). (d) Efficient Channel Attention (ECA). (e) The Convolutional Block Attention Module (CBAM). L denotes sequence length, and W denotes window length.

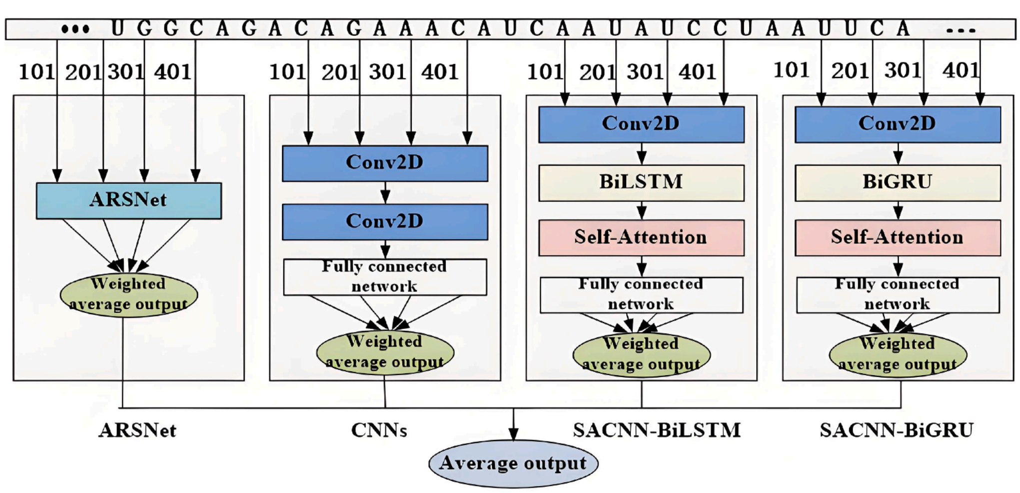

Fig. 2. Architecture of the multi-network fusion baseline (MARSNet-F), which averages predictions from ARSNet, CNN, SACNN-BiLSTM, and SACNN-BiGRU.

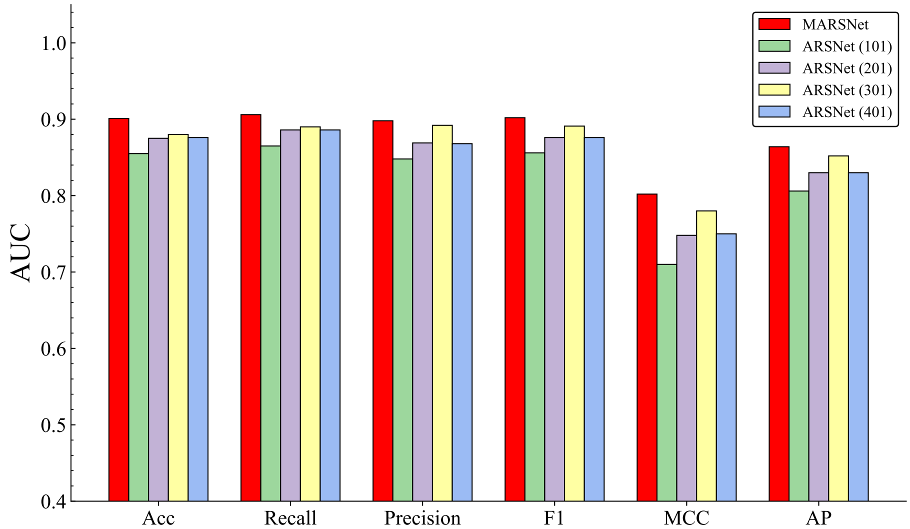

Fig. 3. Analysis of performance indicators of multi-scale windows versus single windows on 24 datasets. MARSNet (red) shows improved average performance and greater stability compared with single-scale variants.

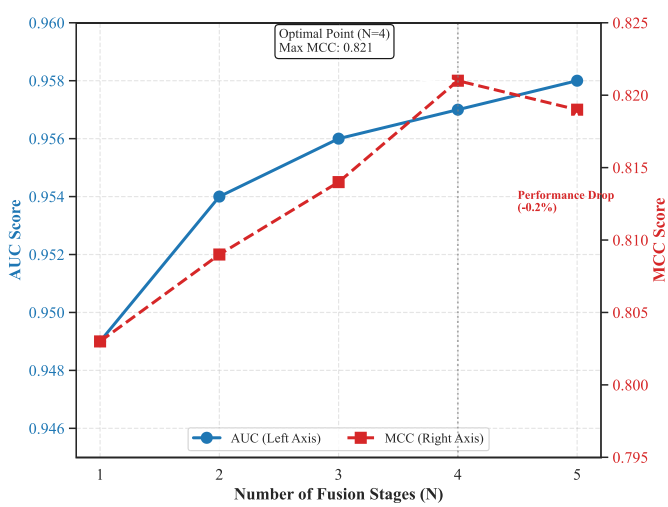

Fig. 4. Performance trend with increasing number of fusion stages (N). Performance improves as N increases and plateaus from $N = 4$ onward.

Table 2
AUC comparison between **MARSNet** ($N = 1$) and ARSNet with single window sizes on 24 datasets. *Note: This table reports the performance of the single-stage multi-scale fusion* ($N = 1$). *The final MARSNet model uses* $N = 4$.

RBPs	MARSNet ($N = 1$)	ARSNet(101)	ARSNet(201)	ARSNet(301)	ARSNet(401)
ALKBH5	0.759	0.666	0.699	0.719	0.712
C17ORF85	0.879	0.731	0.833	0.865	0.828
C22ORF28	0.879	0.800	0.811	0.855	0.850
CAPRIN1	0.958	0.863	0.908	0.953	0.909
AGO2	0.900	0.877	0.875	0.878	0.877
ELAVL1H	0.977	0.962	0.975	0.974	0.974
SRSF1	0.968	0.919	0.942	0.955	0.943
HNRNPC	0.983	0.966	0.975	0.979	0.976
TDP-43	0.956	0.931	0.935	0.951	0.946
TIA1	0.949	0.911	0.928	0.934	0.939
TIAL1	0.949	0.921	0.928	0.942	0.932
AGO1–4	0.946	0.901	0.930	0.923	0.922
ELAVL1B	0.985	0.965	0.982	0.976	0.971
ELAVL1A	0.972	0.949	0.966	0.964	0.974
EWSR1	0.983	0.948	0.971	0.978	0.967
FUS	0.992	0.972	0.989	0.988	0.984
ELAVL1C	0.993	0.987	0.991	0.993	0.990
IGF2BP1–3	0.965	0.926	0.938	0.946	0.942
MOV10	0.958	0.887	0.925	0.939	0.929
PUM2	0.970	0.963	0.965	0.973	0.967
QKI	0.966	0.959	0.960	0.962	0.965
TAF15	0.986	0.970	0.976	0.977	0.983
PTB	0.948	0.929	0.936	0.948	0.942
ZC3H7B	0.959	0.892	0.934	0.951	0.907
Mean	0.949	0.908	0.928	0.938	0.930

(101/201/301/401) and one-hot encoding, as described in Section 2. Hyperparameters were selected empirically with reference to iDeepE and tuned on a validation set (randomly held-out 20% of the training data). Specifically, the first convolutional layer used kernel size $(4, 10)$ with pooling $(1, 3)$ and 16 filters; the second convolutional layer used kernel size $(1, 10)$ with pooling $(1, 3)$ and 32 filters. The residual shrinkage block used kernel size 3. We set the batch size to 128, the hidden units of the fully connected layer to 256, and dropout to 0.25.

3.2. Internal validation and ablation studies

To validate the design of MARSNet, we conducted comprehensive ablation studies to analyze the impact of multi-scale windowing, the number of fusion stages, and the contribution of attention modules.

3.2.1. Effect of multi-scale architecture (windows and fusion stages)

First, we verified the necessity of the multi-scale strategy. Table 2 compares the proposed **MARSNet** ($N = 1$) (single-stage fusion) against ARSNet instances trained with a single fixed-size window. As observed, single-window models show limited performance, often declining when the window becomes overly large. In contrast, by fusing these scales, **MARSNet ($N = 1$)** achieves a mean AUC of **0.949**, demonstrating improved average performance and greater robustness across RBPs. As visually illustrated in Fig. 3, the multi-scale fusion (red bars) consistently yields balanced improvements across all evaluation metrics (Acc, Recall, Precision, F1, MCC, and AP) compared to single-scale variants. While single-window settings may be competitive on specific RBPs, the fused multi-scale model is more stable on average (see Table S3).

We further studied how many fusion stages (N) are optimal. Table 3 and Fig. 4 summarize the performance when varying N from 1 to 5.

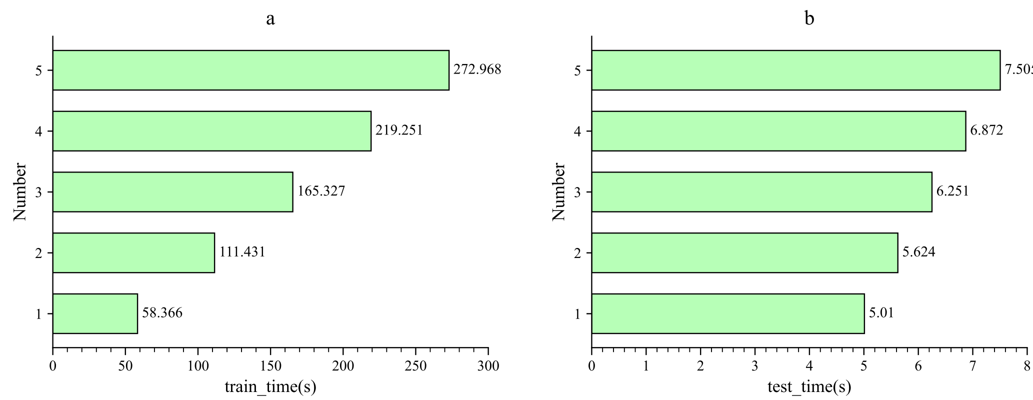

Fig. 5. Time analysis of model structure. (a) Training time. (b) Test time.

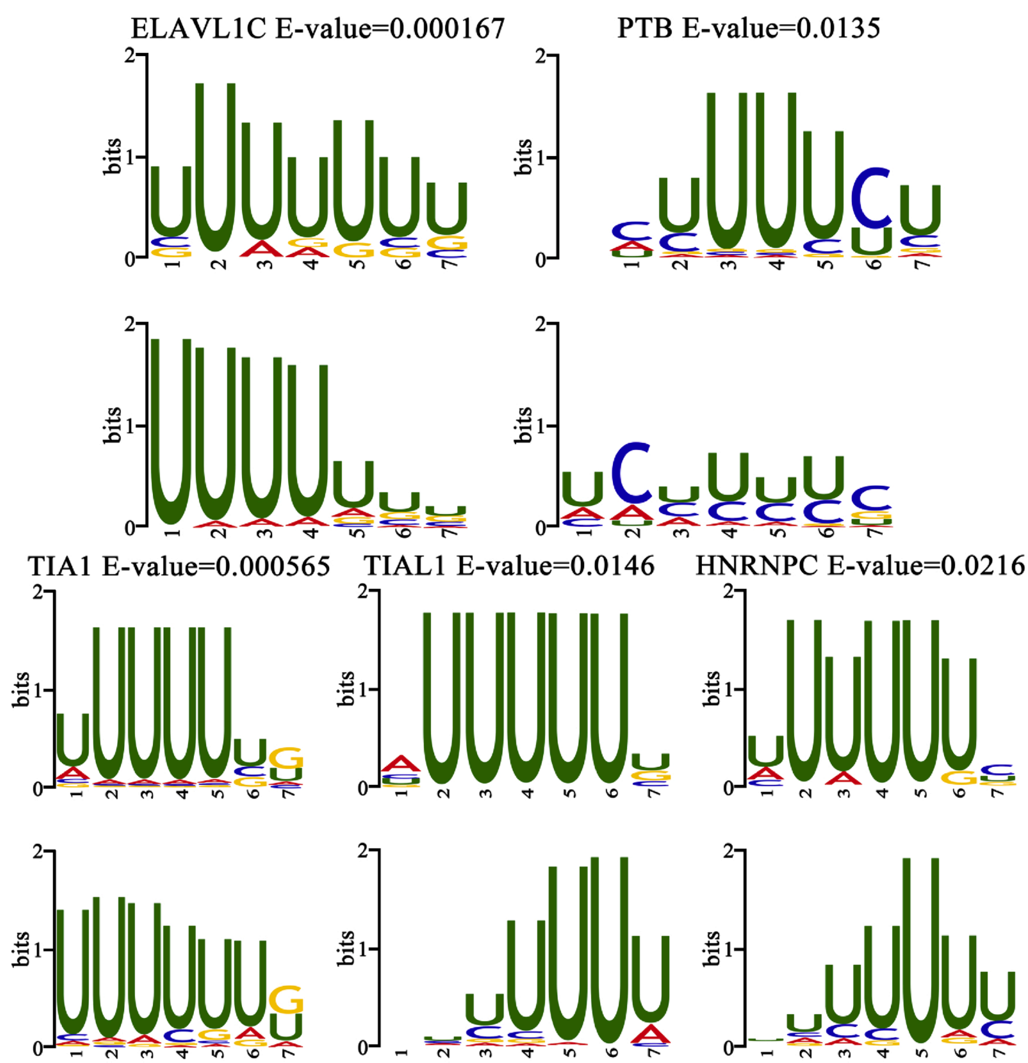

Fig. 6. Binding motifs identified by MARSNet matched to CISBP-RNA database. E-values reported by TOMTOM are shown above each motif logo; the corresponding p/q-values and motif IDs are summarized in Supplementary Table S4.

Performance improves as N increases and plateaus from $N = 4$ onward. Considering the marginal gain versus additional complexity, we set $N = 4$ as the final configuration.

3.2.2. Impact of components and fusion strategy

We validated the contributions of the attention mechanisms and the fusion strategy. As shown in Table 4, removing either ECA or CBAM degrades performance, confirming that these modules effectively capture complementary channel and spatial features. Furthermore, we compared our homogeneous fusion strategy with a heterogeneous baseline, MARSNet-F (Table 5). Results show that repeatedly fusing the optimized feature extractor (MARSNet) yields better performance (AUC 0.957) than simply averaging different model families (MARSNet-F, AUC 0.952).

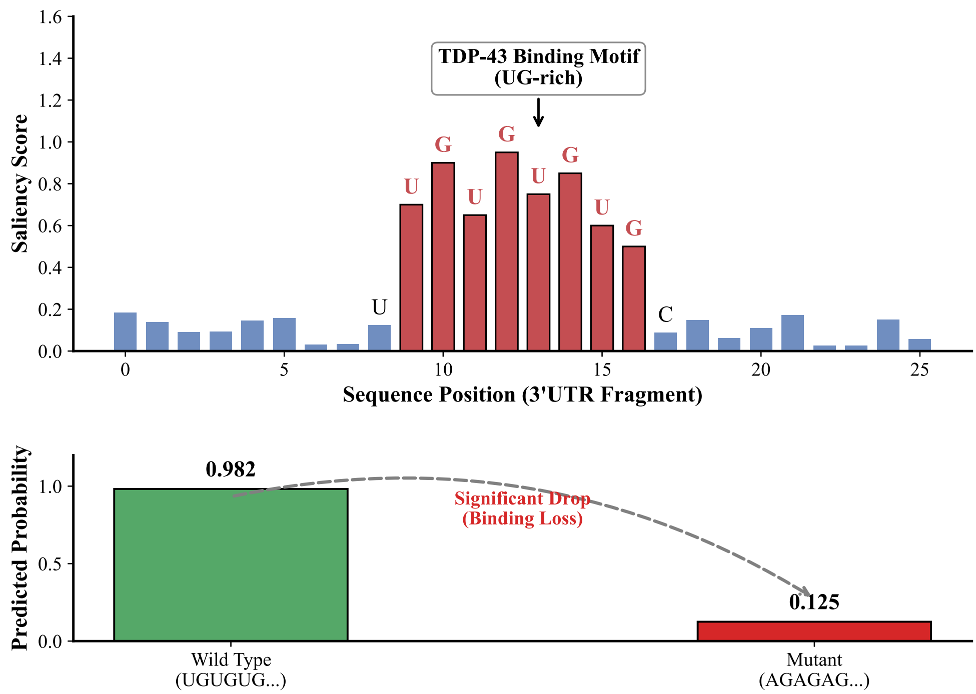

Fig. 7. Case study analysis on *TARDBP*. (A) Saliency map highlighting the UG-rich motif. (B) Mutagenesis results showing probability drop.

Table 3
Performance analysis for different numbers of fusion stages (N).

N	Acc	Recall	Precision	F1	MCC	AP	AUC
1	0.901	0.907	0.898	0.902	0.803	0.864	0.949
2	0.905	0.909	0.902	0.905	0.809	0.868	0.954
3	0.907	0.912	0.904	0.908	0.814	0.872	0.956
4	0.909	0.913	0.907	0.910	0.821	0.875	0.957
5	0.909	0.914	0.907	0.910	0.819	0.874	0.958

Table 4
Ablation study of ECA and CBAM on 24 datasets (mean scores).

Methods	Acc	Recall	Precision	F1	MCC	AP	AUC
No_ECA	0.906	0.910	0.904	0.907	0.811	0.870	0.954
No_CBAM	0.906	0.909	0.905	0.907	0.812	0.872	0.955
MARSNet	**0.909**	**0.913**	**0.907**	**0.910**	**0.821**	**0.875**	**0.957**

Table 5
Fusion strategy comparison on 24 datasets (mean scores).

Method	Acc	Recall	Precision	F1	MCC	AP	AUC
SACNN-BiLSTM	0.879	0.888	0.877	0.882	0.760	0.839	0.930
SACNN-BiGRU	0.877	0.878	0.877	0.877	0.755	0.836	0.929
CNN	0.897	0.900	0.895	0.897	0.794	0.859	0.946
MARSNet ($N=1$)	0.901	0.907	0.898	0.902	0.803	0.864	0.949
MARSNet-F	0.902	0.904	0.902	0.902	0.804	0.866	0.952
MARSNet (Final)	**0.909**	**0.913**	**0.907**	**0.910**	**0.821**	**0.875**	**0.957**

3.3. Computational efficiency

Fusing more modules naturally increases computation. We measured training and inference time on the ALKBH5 dataset (Fig. 5). Panel (a) reports the training time as N increases, and panel (b) reports the corresponding testing time. The figure shows an expected monotonic increase in runtime as more fusion stages are added, which provides a direct trade-off view between accuracy gains and computational cost. Regarding inference latency, while it increases with model complexity (from 55ms at $N=1$ to 230 ms at $N=4$ per 1000 sequences), the final model remains efficient enough for high-throughput screening tasks. To provide a comprehensive comparison against baselines, Table 6 summarizes the parameter size, training speed, and inference latency. The results show that MARSNet can be trained on standard consumer-grade GPUs with a batch size of 128 without exceeding memory limits.

Table 6
Computational complexity analysis. Inference time is measured per 1000 sequences.

Model	Params (M)	Train Time (s/epoch)	Infer. Time (ms/1k)
CNN	0.15	12	45
SACNN-BiLSTM	0.85	35	120
MARSNet ($N=1$)	0.22	18	55
MARSNet ($N=4$)	**0.88**	**75**	**230**

3.4. Biological interpretability

3.4.1. Motif discovery

To connect model predictions with biological insight, we extracted sequence motifs from convolutional filters and matched them to CISBP-RNA using TOMTOM (q-value < 0.05; E-values are shown in Fig. 6, and p/q-values are summarized in Supplementary Table S4).

Fig. 6 shows representative motifs, including U-rich preferences for HNRNPC and TIAL1.

We additionally provide a target-specific interpretability case study for ZC3H7B with motif logos and TOMTOM annotations in the Supplementary Material (Supplementary Fig. S2 and Supplementary Table S4).

3.4.2. Case study: TARDBP mutagenesis

To further validate real-world applicability, we conducted a case study on the *TARDBP* gene. As visualized in Fig. 7(A), the saliency map demonstrates that MARSNet places high importance on the UG-rich region. Technically, the saliency map was derived by extracting and normalizing the spatial attention weights from the CBAM module. The focused attention is consistent with the known biological binding preference of TDP-43 (Tollervey et al., 2011). *In silico* mutagenesis (Fig. 7(B)) shows a drastic drop in binding probability from **0.982** (Wild Type) to

Table 7
Performance comparison (AUC) with state-of-the-art methods on the RBP-24 benchmark. Comparison includes classic baselines and recent models (2023–2025). *Note: Results for PIONet and RMDNet are cited directly from their original publications* (Rashid et al., 2025; Zhang et al., 2025), *which reported performance on the identical RBP-24 benchmark protocol. Statistical significance is assessed by the Wilcoxon signed-rank test across the 24 RBPs; exact (raw and Holm-adjusted) p-values are reported in Supplementary Table S5.* ***n.s.*** *denotes not statistically significant* (*Holm-adjusted* $p \geq .05$).

RBPs (Dataset)	GraphProt (2014)	MCNN (2018)	CRMSNet (2023)	PIONet **(2025)**	RMDNet **(2025)**	MARSNet **(Ours)**
ALKBH5	0.680	0.784	0.732	0.804	0.755	**0.813**
C17ORF85	0.800	0.886	0.876	0.897	0.875	**0.899**
C22ORF28	0.751	0.874	0.871	0.860	0.887	**0.884**
CAPRIN1	0.855	0.948	0.960	0.973	0.926	0.969
AGO2	0.765	0.890	0.884	0.875	0.932	0.912
ELAVL1H	0.955	0.989	0.979	0.984	0.977	0.981
SRSF1	0.898	0.963	0.966	0.968	0.955	**0.975**
HNRNPC	0.952	0.983	0.982	0.987	0.988	0.984
TDP-43	0.874	0.954	0.953	0.962	0.969	0.960
TIAL1	0.861	0.954	0.956	0.941	0.947	**0.957**
TIA1	0.833	0.948	0.942	0.959	0.959	0.955
AGO1–4	0.895	0.940	0.935	0.939	0.942	**0.954**
ELAVL1B	0.959	0.984	0.984	0.987	0.983	**0.987**
ELAVL1A	0.935	0.975	0.972	0.980	0.981	0.975
EWSR1	0.935	0.981	0.983	0.990	0.983	0.986
FUS	0.968	0.991	0.990	0.997	0.994	0.992
ELAVL1C	0.991	0.993	0.997	0.997	0.997	0.992
IGF2BP1–3	0.889	0.970	0.964	0.975	0.961	**0.977**
MOV10	0.863	0.958	0.953	0.963	0.958	**0.969**
PUM2	0.954	0.972	0.971	0.975	0.979	0.973
QKI	0.957	0.974	0.969	0.980	0.982	0.971
TAF15	0.970	0.982	0.981	0.988	0.986	0.986
PTB	0.937	0.950	0.942	0.952	0.971	**0.952**
ZC3H7B	0.882	0.953	0.946	0.966	0.969	**0.968**
Mean	0.890	0.950	0.945	0.954	0.952	**0.957**
Holm-adjusted p	$< 10^{-5}$	$< 10^{-4}$	$< 10^{-3}$	*n.s.*	*n.s.*	–

0.125 (Mutant) when the UG motif is disrupted, confirming MARSNet's sensitivity to functional variations.

3.5. Comparison with state-of-the-art methods

To ensure a fair comparison, all baseline methods were evaluated using the identical training and testing data splits provided by the standard RBP-24 benchmark protocol. We compared MARSNet with baselines ranging from classic GraphProt (2014) to recent deep learning models from 2023–2025: CRMSNet (Pan et al., 2023) (2023), PIONet (2025) (Rashid et al., 2025), and RMDNet (2025) (Zhang et al., 2025).

Table 7 summarizes the AUC scores. MARSNet achieves the highest average AUC of **0.957** and remains competitive with the most recent 2025 models across datasets.

Notably, MARSNet exhibits superior robustness on small-scale datasets. Statistical significance analysis (Wilcoxon signed-rank test) confirms that MARSNet significantly outperforms classic baselines such as GraphProt, MCNN, and CRMSNet ($p < .001$), while remaining competitive with the most recent 2025 models.

4. Conclusion

In this study, we proposed MARSNet, a novel multi-scale convolutional attention residual shrinkage network, to accurately predict RBP binding sites. Addressing the challenges of noise in CLIP-seq data and the limitations of single-scale feature extraction, MARSNet integrates a residual shrinkage mechanism with multi-scale window fusion. The residual shrinkage unit effectively suppresses noise-related signals via soft-thresholding, while the combination of Efficient Channel Attention (ECA) and Convolutional Block Attention Module (CBAM) automatically highlights key sequence features. Extensive experiments on the RBP-24 benchmark demonstrate that MARSNet achieves state-of-the-art performance (mean AUC 0.957), statistically outperforming baselines ranging from classic machine learning models to the latest deep learning methods published in 2025 (e.g., PIONet and RMDNet). Crucially, MARSNet shows exceptional robustness on small-scale datasets where other models often overfit. Beyond numerical superiority, the case study on the *TARDBP* gene validates that MARSNet captures biologically relevant signals, as evidenced by its ability to identify known UG-rich motifs and predict binding loss upon mutation. These results suggest that MARSNet is not only a robust predictor but also a reliable tool for interpreting complex RNA–protein interaction mechanisms.

CRediT authorship contribution statement

Wei Wang: Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization; **Chengyu Xing:** Writing – original draft, Software, Methodology, Data curation; **Zhenxi Sun:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization; **Xianfang Wang:** Writing – review & editing, Funding acquisition; **Guangsheng Wu:** Writing – review & editing, Funding acquisition; **Yun Zhou:** Writing – review & editing, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.neunet.2026.108922.

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